

School of Science and Technology

COURSEWORK

Module Code	: ITEC20261
Module Title	: Advanced Networking
Module Leader	: Dr Evtim Peytchev
Tutor(s)	:
Title	: Cisco Packet Tracer Coursework
Learning outcomes assessed	: K1, K2, K3, S3
Contribution to element	: 50% of the module mark
Date set	: 8 th January 2026
Date of Submission	: 23 rd March 2026
Submission method	: Demo in lab between 23 rd March 2026 18 th April 2026 and ALSO Module Dropbox file submission on NOW
Feedback date	: 18 th April 2026
Feedback method	: Via Module Dropbox
Note	: The usual University penalties apply for late submission and plagiarism. Also, note the university's policy on proof-reading found at Part C Section 15 of the NTU Quality Handbook (13.17 – 13.20). Please consult your student handbook for further details.
NEC	: https://ntu.ac.uk/current_students/resources/student_handbook/appeals/index.html
Writing and maths support	: http://librarybookings.ntu.ac.uk

I. Assessment Requirements

This coursework is designed to increase the student's knowledge in setting up and using Cisco devices. The coursework is set to use the Cisco's Packet Tracer simulation software. Packet Tracer software is advanced networking simulation software and is training and download is available for free on the Cisco web site - <https://www.netacad.com/cisco-packet-tracer>. This is an **individual** coursework.

II. Submission requirements

To pass the coursework you must demonstrated your solution to the lab tutor and submit written report. In the report and during the demo you need to demonstrate your understanding of designing and setting up Cisco devices using Packet Tracer. Your report should be around 5-6 pages including screen shots (if appropriate). Care should be taken to ensure that the report contains correct references to all cited work in an appropriate style, for example, the Harvard Referencing System. You should submit your report to the Dropbox in NOW before the submission deadline.

III. Assessment Scenario/Problem

Advanced Networking Plc., a newly established company in Nottingham, has recently expanded by opening a branch in Derby. While the company has designed its network topology, it requires proper configuration to ensure seamless communication between all host devices across the two branches. They have approached **Nottingham Trent University** with a consulting contract where students in BSc in Advanced Networking will **design, implement/configure, document and ensure seamless connectivity** to meet the company's specific requirements.

This document describes the **current state of the network topology** as well as the **security requirements** of Advanced Networking Plc. Finally, it describes the **final deliverables** for this assignment.

Initial Topology

The company has **headquarters** in **Nottingham** and a branch in **Derby**. **Figure 1** depicts the current topology of the company's network.

The network is **structured** as follows:

- **PUBLIC NETWORK:** The public network lies outside Advanced Networking Plc's administrative domain and is managed by an Internet Service Provider (ISP). Communication between Advanced Networking Plc.'s Nottingham and Derby branches occurs through the ISP's public IP networks, specifically 209.165.100.0/30 and 209.165.100.4/30.

The ISP's internal infrastructure utilizes private IP addressing and includes the following key components:

1. HTTPS Server: Provides secure web services.
2. DHCP Server: Dynamically assigns IP addresses to devices connected to the network.

Use the network address 192.168.1.0/24 to assign IP addresses to the router interface and the two servers. For the servers, the only configuration required is to assign the following: IP Address, Subnet Mask and Default Gateway.

Ensure that all addresses are within the specified network range.
This configuration ensures connectivity and access to essential public resources while maintaining network flexibility for Advanced Networking Plc.

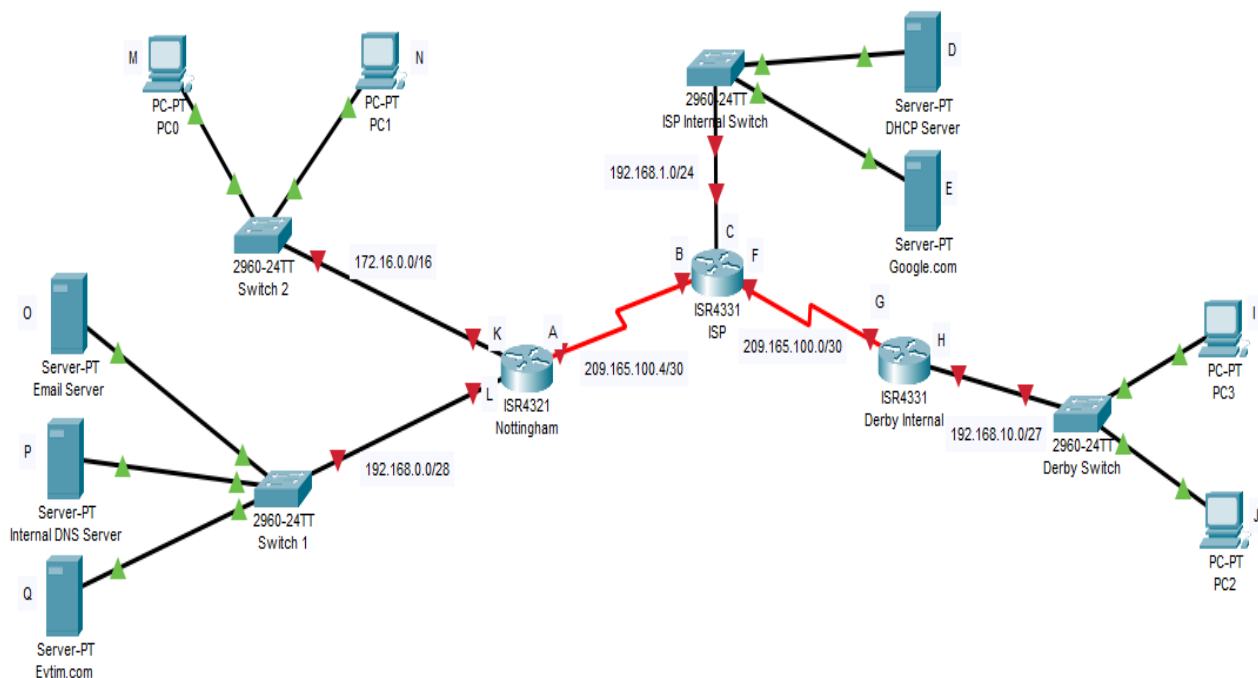


Figure 1

You required to perform the following:

Design the Network Topology (15%)

Design the network topology using Packet Tracer. The name of the PT file must be your SID number, i.e. N1234567.pka.

Configure the Network Topology (50%)

Use the network address provided in **Figure 1** to plan and configure the network.

Table 1: Complete and show the IP addresses and masks used configured in each device interface

Interface	IP address	Subnet mask
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		
K		
L		
M		
N		
O		
P		
Q		

Table 1. Advanced Networking Plc IP address scheme

Testing of the Network Topology (25%)

Finally, you must provide a test plan. This plan should include **screenshots** that demonstrate successful connectivity between the two branches and the ISP. The screenshots should include connectivity between

- M and I interfaces
- J and E interfaces
- N and Q interfaces

Overall Document Quality (10%)

Overall Quality and structure of the report must be appropriate.

IV. Marking Rubric for Assignment Element (Case Study)

Component	Description	Value (%)	Submission
1	Design - Demo	15%	PKA file - Demo
2	Configuration - Demo	50%	PKA file - Demo
3	Test Plan	25%	Report
4	Overall Document Quality	10%	Report

V. Feedback Opportunities

Formative (Whilst you're working on the coursework)

You will be given the opportunity to receive informal verbal feedback from your tutor regarding your coursework development during the lab sessions.

Summative (After you've submitted the coursework)

You will receive specific feedback regarding your coursework submission together with your awarded mark when it is returned to you. Clearly, feedback provided with your coursework is only for developmental purposes so that you can improve for the next assessment or subject-related module.

V. Resources that may be useful

Referencing styles please use Harvard as detailed [here](#)

Guide to planning your time [here](#) and an automated planner [here](#)

[Guidance on avoiding cheating is here](#)

Remember to use Outlook or physical calendars to block out time between lectures and labs to work on this coursework.

VI. Moderation

The Moderation Process

All assessments are subject to a two-stage moderation process. Firstly, any details related to the assessment (e.g., clarity of information and the assessment criteria) are considered by an independent person (usually a member of the module team). Secondly, the grades awarded are considered by the module team to check for consistency and fairness across the cohort for the piece of work submitted.

Grading Descriptors

Class	Scale	General Characteristics
FIRST (Excellent)	Exceptional 1st	Exceptional breadth and depth of knowledge and understanding of the area of study; evidence of extensive and appropriate selection and critical evaluation/synthesis/analysis and of reading/research beyond the prescribed range, in both breadth and depth, to advance work/direct arguments; exceptional demonstration of relevant skills; excellent communication; performance deemed to be beyond expectation.
	High 1st	Outstanding/excellent knowledge and understanding of the area of study as the student is typically able to go beyond what has been taught (particularly for a mid/high 1st) ; evidence of extensive and appropriate selection and critical evaluation/synthesis/analysis of reading/research beyond the prescribed range , to advance work/direct arguments; excellent demonstration of relevant skills; excellent communication; performance deemed beyond expectation of the level.
	Mid 1st	
	Low 1st	
UPPER SECOND (Very good)	High 2.1	Very good knowledge and understanding of the area of study as the student is typically able to relate facts/concepts together with some ability to apply to known/taught contexts ; evidence of appropriate selection and evaluation of reading/research, some beyond the prescribed range, may rely on set sources to advance work/direct arguments; demonstrates autonomy in approach to learning; very good demonstration of relevant skills; strong communication skills.
	Mid 2.1	
	Low 2.1	
LOWER SECOND (Good)	High 2.2	Good knowledge and understanding of the area of study balanced towards the descriptive rather than analytical ; evidence of appropriate selection and evaluation of reading/research but generally reliant on set sources to advance work/direct arguments; good demonstration of relevant skills, though may be limited in range; communication shows clarity but structure may not always be coherent.
	Mid 2.2	
	Low 2.2	
THIRD (Sufficient)	High 3 rd	Knowledge and understanding is sufficient to deal with terminology, basic facts and concepts but fails to make meaningful synthesis; some ability to select and evaluate reading/research however work may be more generally descriptive; strong reliance on available support set sources to advance work; arguments may be weak or poorly constructed; adequate demonstration of relevant skills over a limited range; communication/presentation is generally competent but with some weaknesses.
	Mid 3 rd	
	Low 3 rd	
FAIL (Insufficient)	Marginal Fail	Insufficient knowledge and understanding of the area of study; some ability to select and evaluate reading/research however work is more generally descriptive; fails to address some aspects of the brief; a limited use of sources to advance work; arguments may be weak/poor or weakly/poorly constructed; demonstration of relevant skills over a reduced range; communication shows limited clarity, poor presentation, structure may not be coherent.
	Mid Fail	Highly insufficient knowledge or understanding of the area of study; understanding is typically at the word level with facts being reproduced in a disjointed or decontextualised manner ; fails to address the outcomes addressed by the brief; typically ignores important sources in development of work and data/evidence inappropriately used; weak technical and practical competence hampers ability to demonstrate/communicate achievement of outcomes.
	Low Fail	
ZERO	Zero	Work of no merit OR absent, work not submitted, penalty in some misconduct cases.